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Optimizing Jupyter Notebook: Tips, Tricks, and nbextensions

Jupyter Notebooks are a web-based and interactive tool that the machine learning and data science community uses a lot. They are used for quick testing, as a reporting tool or even as highly sophisticated learning materials in online courses.

So here in this blog, I'm going to list down a few of the shortcuts, magic commands and nbextensions.

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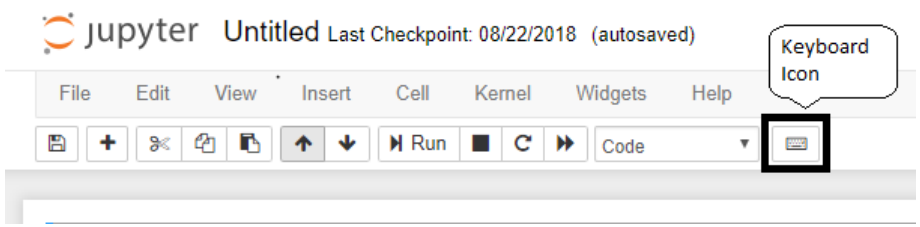
Shortcuts

Press `Ctrl+Shift+p` or click on the small keyboard icon from the menu bar to get a list of command palette



Shortcuts in both command and edit modes:

- `Shift + Enter` run the current cell, select below
- `Ctrl + Enter` run selected cells
- `Alt + Enter` run the current cell, insert below
- `Ctrl + S` save and checkpoint

Shortcuts when in edit mode:

- `Esc` take you into command mode
- `Tab` code completion or indent
- `Shift + Tab` tooltip
- `Ctrl + ]` indent
- `Ctrl + [` dedent

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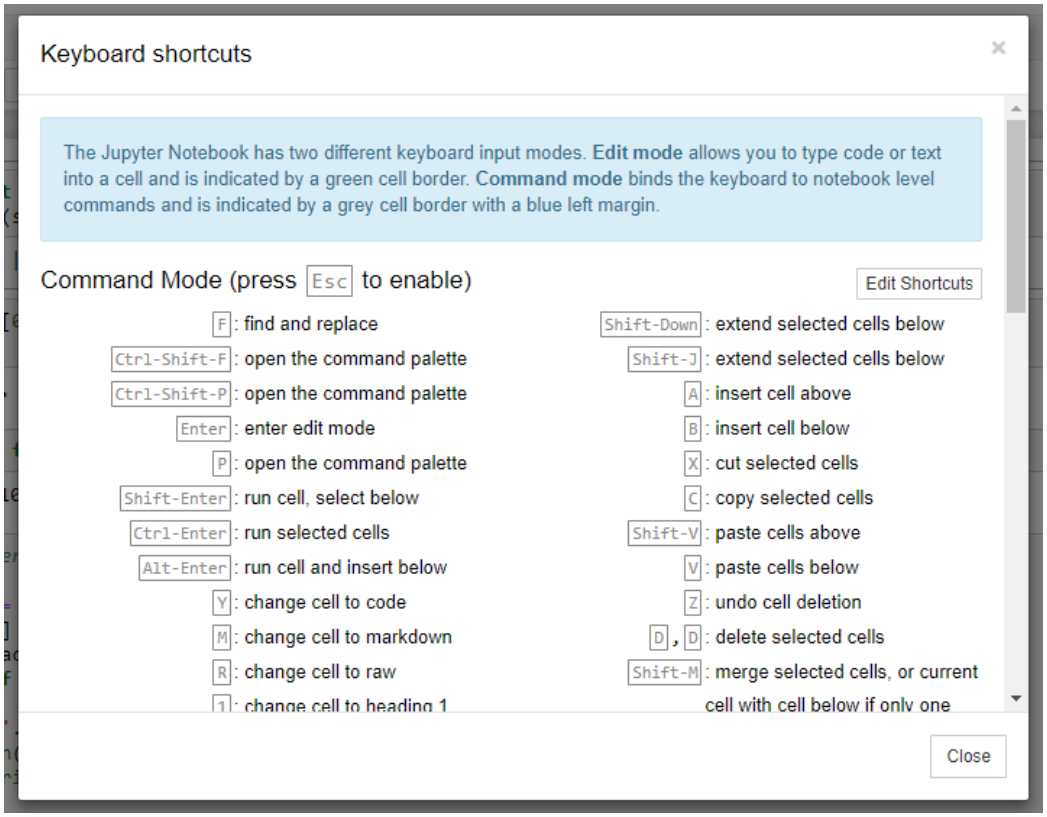
- `Ctrl + Shift + Z` or `Ctrl + Y` redo
- `Ctrl + Home` go to cell start
- `Ctrl + End` go to the cell end
- `Ctrl + Left` go one word left
- `Ctrl + Right` go one word right

Once in command mode, press `H` (help) to get a list of keyboard shortcuts:



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- Scroll up and down your cells with your `Up` and `Down` keys.
- Press `A` or `B` to insert a new cell above or below the active cell.
- `M` will transform the active cell to a Markdown cell.
- `Y` will set the active cell to a code cell.
- `X` will cut the selected cell
- `C` will copy the selected cell
- `V` will paste the cell which is being copied/cut
- `Shift + V` paste cells above
- `S` will save the notebook
- `F` will find/replace
- `O` will toggle output
- `D + D` (`D` twice) will delete the active cell.
- `Z` will undo cell deletion.
- For selecting multiple cells at once, hold `Shift` and press `Up` or `Down`
- `Shift + Space` scroll notebook up
- `Space` scroll notebook down

With multiple cells selected:

- Press `Shift + M` to merge your selection
- To split the active cell at the cursor, press `Ctrl + Shift + -` in edit mode

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Copying and pasting cells BETWEEN notebooks:

- Notebook 1: — Select multiple cells by holding down Shift and hit `Ctrl+c` to copy
- Notebook 2: — Hit Esc to enter Command mode `Ctrl + v` to paste

Installing packages in the current kernel using `! pip install <package>`

```
In [1]: ! pip install pandas
```

Using Python variables in shell commands by prepending a `$` symbol:

```
In [46]: x = "Hello World!"
         !echo $x

         Hello World!
```

. . .

Magic Commands

Magic Commands are shortcuts that significantly extend a notebook's capabilities

Sharing a code sample from your notebook:

1. Use the `%pastebin` magic function to select a range of cells

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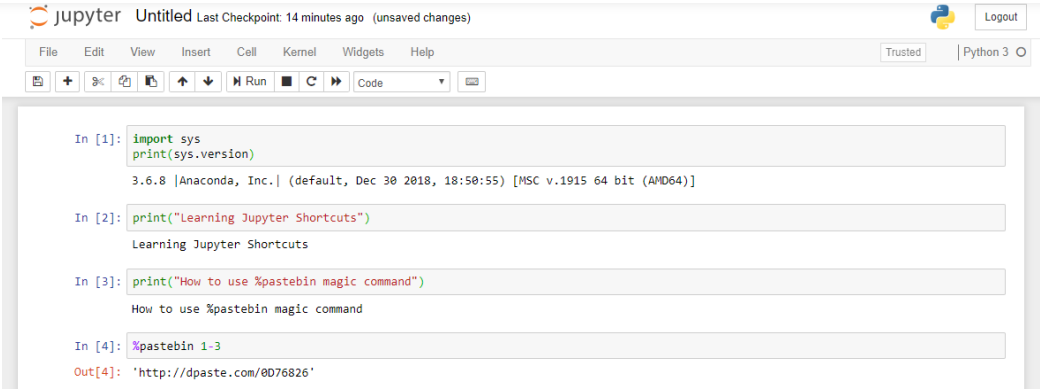

Note- This link gets expired in 7 days

To have a list of defined variables, use `%whos` or `%who_ls`

```
In [1]: x, y = "Hello", "World"

In [2]: %whos

        Variable   Type    Data/Info
        ---------------------------
        x          str     Hello
        y          str     World

In [3]: %who_ls

Out[3]: ['x', 'y']
```

- `%whos` displays the variable type plus some extra info: size, contents, etc.
- `%who_ls` only displays the variables name

Working with external files within Jupyter Notebook:

- `%pycat file.py` ➡ opens script in the pager

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- `%run file.ipynb` ➞ runs notebook
- `%notebook filename` ➞ exporting the current IPython history to a notebook file

Getting, setting or listing environment variables:

- `%env` ➞lists all environment variables
- `%env var` ➞ get value for var
- `%env var val` ➞ set value for var

Running commands in the shell:

`%system` ➞ to use shell (mostly used to get current directory, date, etc)

```
In [43]: %system time
Out[43]: ['The current time is:  2:26:09.12', 'Enter the new time: ']
```

Autosaving notebook to its checkpoint using `%autosave` :

```
In [47]: %autosave 120
         Autosaving every 120 seconds
```

Autosaving every 120 seconds (2 minutes)

Executing different languages:

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```
In [3]: %%HTML
        This is <i>HTML</i> code!
```

This is *HTML* code!

- `%%perl` → to execute Perl in a subprocess
- `%%javascript` or `%%js` → to execute the block of Javascript code
- `%%python3` → to execute code with python3 in a subprocess
- `%%ruby` → to execute Ruby code

Other magic commands:

- `%history` → print input history
- `%lsmagic` → list currently available magic functions
- `%magic` → print information about the magic function system
- `%matplotlib` → setting up matplotlib to work interactively
- `%pwd` → returns current working directory
- `%quickref` → shows a quick reference sheet
- `%time` → time execution of a Python statement or expression (it can be used both as a line and cell magic)

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nbextensions

The benefits of this extension are that it changes the defaults

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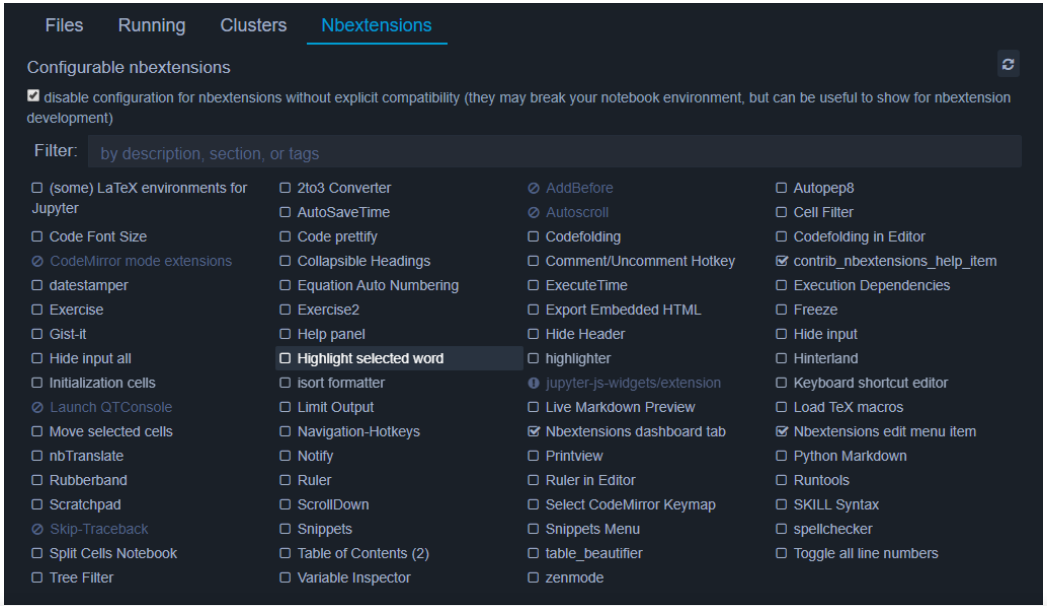
```
conda install -c conda-forge jupyter_contrib_nbextensions
conda install -c conda-forge jupyter_nbextensions_configurator
```

Alternatively, you can also install nbextensions using pip:

- `pip install jupyter_contrib_nbextensions`
- Run `pip show jupyter_contrib_nbextensions` to find where notebook extensions are installed
- Run `jupyter contrib nbextensions install` to install the new extension

After installing, restart the Jupyter notebook, and you can observe a new tab *Nbextensions* added to the menu:



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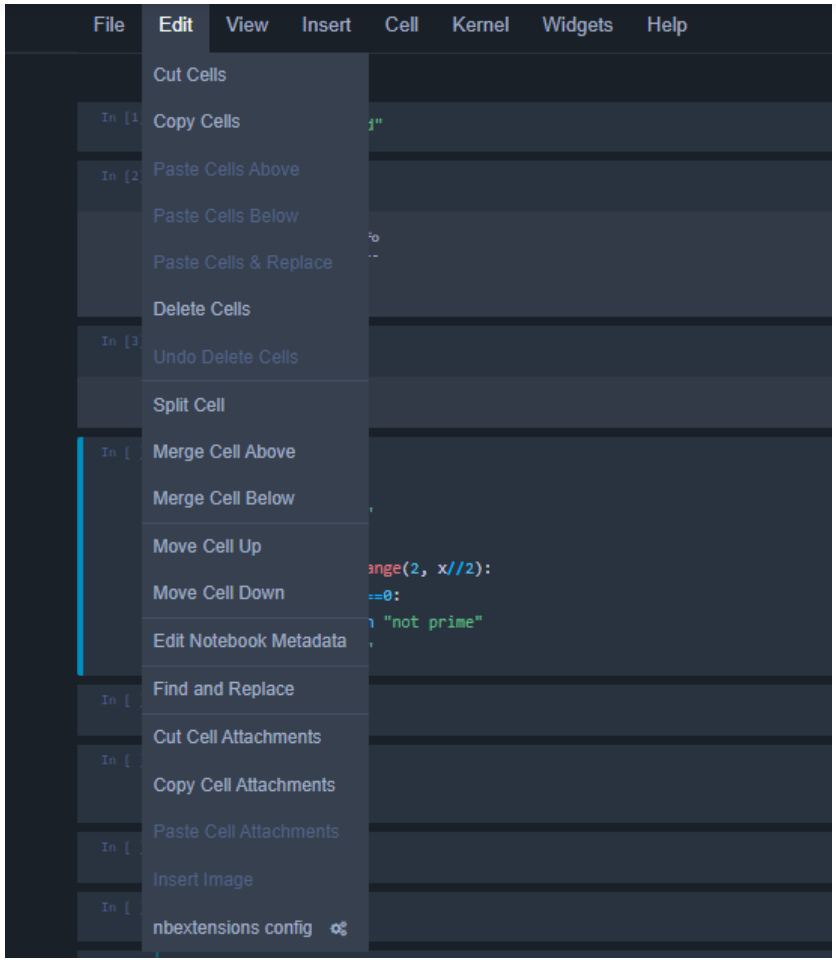
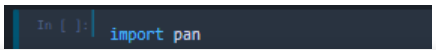


Now, let's see a few of the nbextensions:

1. **Hinterland** - It enables code autocompletion menu for every keypress in a code cell, instead of only enabling it with tab



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2. **Split Cells Notebook** - Enable split cells in Jupyter notebooks

Enter command mode (`Esc`), use `Shift + s` to toggle the current cell to either a split cell or full width.

3. **Table of Contents** - The toc extension enables to collect all running headers and display them in a floating window, as a sidebar or with a navigation menu. The extension is also draggable, resizable, collapsable, dockable and features automatic numerotation with unique links ids, and an optional toc cell.

4. **Autopep8 -** Use kernel-specific code to reformat/prettify the contents of code cells

5. **Snippets** - Adds a drop-down menu to insert snippet cells into the current notebook.

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Jupyter Notebook Theme

We can convert to Dark mode from default theme of Jupyter Notebook.

For achieving this, we need to install `jupyterthemes` :

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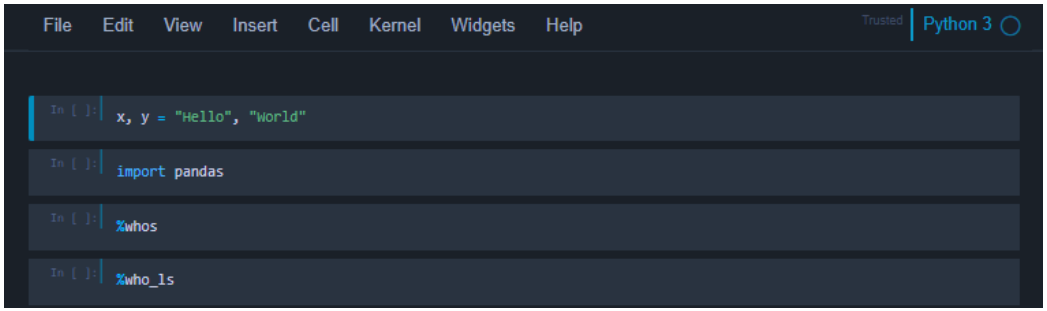
`jt -l` will give the list of available themes

```
In [1]: ! jt -l

Available Themes:
   chesterish
   grade3
   gruvboxd
   gruvboxl
   monokai
   oceans16
   onedork
   solarizedd
   solarizedl
```

`jt -t <theme name>` will change the theme. Let's try changing it to the dark mode by using the *chesterish* theme.

```
In [ ]: ! jt -t chesterish
```



`jt -r` will restore it to the default theme

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There are many other things we can do with Jupyter notebooks that we haven't

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